

SRI VASAVI COLLEGE, SFW, ERODE

C++ MCQ with ANSWERS

Oops (40), C++ Fundamentals(30), Constants & Data types(30), Decision Making (20), Operators, Functions(20), Arrays(30), Structures(10) & Strings, Pointer(20), File(30)

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OOPS CONCEPTS

1. Wrapping data and its related functionality into a single entity is known as _____
a) Abstraction **b) Encapsulation** c) Polymorphism d) Modularity
2. How structures and classes in C++ differ?
a) In Structures, members are public by default whereas, in Classes, they are private by default
b) In Structures, members are private by default whereas, in Classes, they are public by default
c) Structures by default hide every member whereas classes do not
d) Structures cannot have private members whereas classes can have
3. What does polymorphism in OOPs mean?
a) Concept of allowing overriding of functions b) Concept of hiding data
c) Concept of keeping things in different modules/files
d) Concept of wrapping things into a single unit
4. Which concept allows you to reuse the written code?
a) Encapsulation b) Abstraction **c) Inheritance** d) Polymorphism
5. Which of the following explains Polymorphism?
a) int func(int, int); float func1(float, float); b) int func(int); int func(int);
c) int func(float); float func(int, int, char); d) int func(); int new_func();
6. Which of the following shows multiple inheritances?
a) A->B->C b) A->B; A->C **c) A,B->C** d) B->A
7. How access specifiers in Class helps in Abstraction?
a) They does not helps in any way
b) They allows us to show only required things to outer world
c) They help in keeping things together d) Abstraction concept is not used in classes

8. C++ is _____

- a) procedural programming language
- b) object oriented programming language
- c) functional programming language
- d) both procedural and object oriented programming language**

9. What does modularity mean?

- a) Hiding part of program
- b) Subdividing program into small independent parts**
- c) Overriding parts of program
- d) Wrapping things into single unit

10. Which of the following feature of OOPs is not used in the following C++ code?

```
class A{
    int i;
    public:
    void print() {cout << "hello" << i;}
}
class B : public A{
    int j;
    public:
    void assign (int a ) {k = a;}
}
```

- a) Abstraction
- b) Encapsulation
- c) Inheritance
- d) Polymorphism**

11. Which of the following class allows to declare only one object of it?

- a) Abstract class
- b) Virtual class
- c) Singleton class**
- d) Friend class

12. Which of the following is not a type of Constructor?

- a) Friend constructor**
- b) Copy constructor
- c) Default constructor
- d) Parameterized constructor

13. Which of the following is correct?

- a) Base class pointer object cannot point to a derived class object
- b) Derived class pointer object cannot point to a base class object**
- c) A derived class cannot have pointer objects
- d) A base class cannot have pointer objects

14. Out of the following, which is not a member of the class?

- a) Static function
- b) Friend function**
- c) Constant function
- d) Virtual function

15. What is the other name used for functions inside a class?

- a) Member variables
- b) Member functions**
- c) Class functions
- d) Class variables

16. Which of the following cannot be a friend?

- a) Function
- b) Class
- c) Object**
- d) Operator function

17. Why references are different from pointers?

- a) A reference cannot be made null
- b) A reference cannot be changed once initialized**
- c) No extra operator is needed for dereferencing of a reference
- d) All of the mentioned

18. Which of the following provides a programmer with the facility of using object of a class inside other classes? a) Inheritance **b) Composition** c) Abstraction d) Encapsulation

19. How many types of polymorphism are there in C++?
a) 1 **b) 2** c) 3 d) 4

20. How run-time polymorphisms are implemented in C++?
a) Using Inheritance b) Using Virtual functions
c) Using Templates **d) Using Inheritance and Virtual functions**

21. How compile-time polymorphisms are implemented in C++?
a) Using Inheritance b) Using Virtual functions
c) Using Templates d) Using Inheritance and Virtual functions

22. Which of the following is an abstract data type?
a) int b) float **c) class** d) string

23. Which concept means the addition of new components to a program as it runs?
a) Data hiding b) Dynamic binding **c) Dynamic loading** d) Dynamic typing

24. Which of the following explains the overloading of functions?
a) Virtual polymorphism b) Transient polymorphism
c) Ad-hoc polymorphism d) Pseudo polymorphism

25. Which of the following approach is used by C++?
a) Top-down **b) Bottom-up** c) Left-right d) Right-left

26. Which operator is overloaded for a cout object?
a) >> **b) <<** c) < d) >

27. Which of the following cannot be used with the virtual keyword?
a) Class b) Member functions **c) Constructors** d) Destructors

28. Which concept is used to implement late binding?
a) Virtual functions b) Operator functions
c) Constant functions d) Static functions

29. Which of the following is correct?
a) C++ allows static type checking b) C++ allows dynamic type checking.
c) C++ allows static member function to be of type const.
d) C++ allows both static and dynamic type checking

30. Which of the following supports the concept that reusability is a desirable feature of a language?
a) It reduces the testing time b) It reduces maintenance cost
c) It decreases the compilation time **d) It reduced both testing and maintenance time**

31. Which of the following is a static polymorphism mechanism?

a) Function overloading

b) Operator overloading

c) Templates

d) All of the mentioned

32. Which of the following is true?

I) All operators in C++ can be overloaded.

II) The basic meaning of an operator can be changed.

a) I only

b) II only

c) Both I and II

d) Neither I nor II

33. Which of the following is not a type of inheritance?

a) Multiple

b) Multilevel

c) Distributive

d) Hierarchical

34. What happens if a class does not have a name?

a) It will not have a constructor

b) It will not have a destructor

c) It is not allowed

d) It will neither have a constructor or destructor

35. Which of the following statement is true?

I) In Procedural programming languages, all function calls are resolved at compile-time

II) In Object Oriented programming languages, all function calls are resolved at compile-time

a) I only

b) II only

c) Both I and II

d) Neither I nor II

36. Which members are inherited but are not accessible in any case?

a) Private

b) Public

c) Protected

d) Both private and protected

37. Which of the following is correct?

a) Friend functions can access public members of a class

b) Friend functions can access protected members of a class

c) Friend functions can access private members of a class

d) All of the mentioned

38. Which of the following is correct in C++?

a) Classes cannot have protected data members

b) Structures can have member functions

c) Class members are public by default

d) Structure members are private by default

39. Which of the following is used to make an abstract class?

a) By using virtual keyword in front of a class declaration

b) By using an abstract keyword in front of a class declaration

c) By declaring a virtual function in a class

d) By declaring a pure virtual function in a class

40. Which of the following is correct?

a) A class is an instance of its objects

b) An object is an instance of its class

c) A class is an instance of the data type that the class have

d) An object is an instance of the data type of the class



Fundamentals

1. Who created C++?

a) Bjarne Stroustrup

b) Dennis Ritchie

c) Ken Thompson

d) Brian Kernighan

2. C++ Language developed at _____?

a) AT & T's Bell Laboratories of USA in 1972

b) Sun Microsystems in 1973

c) Cambridge University in 1972

d) AT & T's Bell Laboratories of USA in 1979

3. Which one of the following is a keyword?

a) Size b) Key c) Jump d) Switch

4. _____ is the smallest individual unit in a program.

a) Variable b) Control c) Character d) Token

5. Which of the following is the correct syntax of including a user defined header files in C++?

a) #include <userdefined.h> b) #include <userdefined>
c) **#include "userdefined"** d) #include [userdefined]

6. Which of the following is a correct identifier in C++?

a) 7var_name b) 7VARNAME c) **VAR_1234** d) \$var_name

7. Which of the following is called address operator?

a) * b) **&** c) _ d) %

8. C++ is _____ type of programming language.?

a) **Object Oriented** b) Procedural c) Bit level language d) Functional

9. Which of the following is used for comments in C++?

a) // comment b) /* comment */ c) **both // comment or /* comment */** d) // comment */

10. What are the actual parameters in C++?

a) **Parameters with which functions are called**
b) Parameters which are used in the definition of a function
c) Variables other than passed parameters in a function
d) Variables that are never used in the function

11. What are the formal parameters in C++?

a) Parameters with which functions are called
b) **Parameters which are used in the definition of the function**
c) Variables other than passed parameters in a function
d) Variables that are never used in the function

12. Which function is used to read a single character from the console in C++?

a) **cin.get(ch)** b) getline(ch) c) read(ch) d) scanf(ch)

13. Which function is used to write a single character to console in C++?

a) **cout.put(ch)** b) cout.putline(ch) c) write(ch) d) printf(ch)

14. What is a constant that contains a single character enclosed within single quotes?

a) Numeric b) Fixed c) **Character** d) Floating Point

15. A C++ code line ends with _____

a) **A Semicolon (;)** b) A Fullstop(.) c) A Comma (,) d) A Slash (/)

16. What are the escape sequences?

a) **Set of characters that convey special meaning in a program**
b) Set of characters that whose use are avoided in C++ programs

- c) Set of characters that are used in the name of the main function of the program
- d) Set of characters that are avoided in cout statements

17. Which of the following escape sequence represents carriage return?

- a) \r b) \n c) \n\r d) \c

18. Which of the following escape sequence represents tab?

- a) \t b) \tr c) \b d) \a

19. Which of the following is called insertion/put to operator?

- a) >> b) << c) > d) <

20. Which of the following is called extraction/get from operator?

- a) << b) >> c) > d) <

21. A language which has the capability to generate new data types are called _____

- a) **Extensible** b) Overloaded c) Encapsulated d) Reprehensible

22. Which of the following is not a fundamental type is not present in C but present in C++?

- a) int b) float c) **bool** d) void

23. What is the size of a boolean variable in C++?

- a) **1 bit** b) 1 byte c) 4 bytes d) 2 bytes

24. Which of the following is C++ equivalent for scanf()?

- a) **cin** b) cout c) print d) input

25. Which of the following is C++ equivalent for printf()?

- a) cin b) **cout** c) print d) input

26. Which of the following is an exit-controlled loop?

- a) for b) while c) **do-while** d) all of the mentioned

27. Which of the following is the correct difference between cin and scanf()?

- a) both are the same b) **cin is a stream object whereas scanf() is a function**
- c) scanf() is a stream object whereas cin is a function
- d) cin is used for printing whereas scanf() is used for reading input

28. Which of the following is an entry-controlled loop?

- a) for b) while c) do-while d) **both while and for**

29. In which part of the for loop termination condition is checked?

- for(I;II;III) a) I b) **II** c) III d) IV

30. Which of the following is the scope resolution operator?

- a) . b) * c) **::** d) ~

Constants & Data Types

1. The constants are also called as

a) Const b) preprocessor **c) literals** d) none of these

2. What are the parts of the literal constants?

a) integer numerals b) floating-point numerals
c) strings and boolean values **d) all of the mentioned**

3. What is the size of `wchar_t` in C++?

a) 2 b) 4 c) 2 or 4 **d) Based on the number of bits in the system**

4. Pick the odd one out.

a) array type b) character type c) boolean type d) integer type

5. Which data type is used to represent the absence of parameters?

a) int b) short **c) void** d) float

6. What does '\a' escape code represent?

a) alert b) backslash c) tab d) form feed

7. How the constants are declared?

a) const keyword b) #define preprocessor **c) both a and b** d) None of these

8. Which type is best suited to represent the logical values?

a) integer **b) Boolean** c) character d) float

9. Identify the user-defined types from the following?

a) enumeration b) classes **c) both enumeration and classes** d) int

10. Which of the following statements are true?

int f(float)

a) f is a function taking an argument of type int and returning a floating point number

b) f is a function taking an argument of type float and returning an integer

c) f is a function of type float d) f is a function of type int

11. The value 132.54 can be represented using which data type?

a) double b) void c) int d) bool

12. When a language has the capability to produce new data type mean, it can be called as

a) overloaded **b) extensible** c) encapsulated d) reprehensible

13. Pick the odd one out.

a) integer, character, boolean, floating b) enumeration, classes
c) integer, enum, void d) arrays, pointer, classes

14. How many characters are specified in the ASCII scheme?

a) 64 **b) 128** c) 256 d) 24

15. Given the variables p, q are of char type and r, s, t are of int type. Select the right statement?

1. $t = (r * s) / (r + s);$ 2. $t = (p * q) / (r + s);$
a) 1 is true but 2 is false b) 1 is false and 2 is true

c) both 1 and 2 are true

d) both 1 and 2 are false

16. Which of the following belongs to the set of character types?

a) char

b) wchar_t

c) only a

d) both wchar_t and char

17. How do we represent a wide character of the form wchar_t?

a) L'a'

b) l'a'

c) L[a]

d) la

18. In C++, what is the sign of character data type by default?

a) Signed

b) Unsigned

c) Implementation dependent

d) Unsigned Implementation

19. Is the size of character literals different in C and C++?

a) Implementation defined

b) Can't say

c) Yes, they are different

d) No, they are not different

20. Suppose in a hypothetical machine, the size of char is 32 bits. What would sizeof(char) return?

a) 4

b) 1

c) Implementation dependent

d) Machine dependent

21. What constant defined in <climits> header returns the number of bits in a char?

a) CHAR_SIZE

b) SIZE_CHAR

c) BIT_CHAR

d) CHAR_BIT

22. The size_t integer type in C++ is?

a) Unsigned integer of at least 64 bits

b) Signed integer of at least 16 bits

c) Unsigned integer of at least 16 bits

d) Signed integer of at least 64 bits

23. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int x = -1;
    unsigned int y = 2;

    if (x > y) {
        cout << "x is greater: " ;
    }
    else
    {
        cout << "y is greater: " ;
    }
    return 0;
}
```

a) x is greater

b) y is greater

c) implementation defined

d) arbitrary

24. Which of these expressions will return true if the input integer v is a power of two?

a) (v | (v + 1)) == 0;

b) (~v & (v - 1)) == 0;

c) (v | (v - 1)) == 0;

d) (v & (v - 1)) == 0;

25. What is the value of the following 8-bit integer after all statements are executed?

1. int x = 1;

2. x = x << 7;

3. x = x >> 7;

a) 1

b) -1

c) 127

d) Implementation defined

26. Which of these expressions will make the rightmost set bit zero in an input integer x?

- a) $x = x \mid (x-1)$ b) $x = x \& (x-1)$ c) $x = x \mid (x+1)$ d) $x = x \& (x+2)$

27. Which of these expressions will isolate the rightmost set bit?

- a) $x = x \& (\sim x)$ b) $x = x \wedge (\sim x)$ c) $x = x \& (-x)$ d) $x = x \wedge (-x)$

28. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int x = 8;
    cout << "ANDing integer 'x' with 'true' : " << x && true;
    return 0;
}
```

- a) ANDing integer 'x' with 'true' :8 b) ANDing integer 'x' with 'true' :0
c) ANDing integer 'x' with 'true' :1 d) ANDing integer 'x' with 'true' :9

29. 0946, 786427373824, 'x' and 0X2f are _____ and _____ literals respectively.

- a) decimal, character, octal, hexadecimal b) octal, hexadecimal, character, decimal
c) hexadecimal, octal, decimal, character d) octal, decimal, character, hexadecimal

30. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int i = 3;
    int l = i / -2;
    int k = i % -2;
    cout << l << k ;
    return 0;
}
```

- a) compile time error b) -1 1 c) 1 -1 d) implementation defined

Decision making & Loops

1. How many sequences of statements present in C++?

- a) 4 b) 3 c) 5 d) 6

2. Decision Control statements in C++ can be implemented using

- a) if b) if-else c) Conditional Operator d) All of the above

3. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
```



```
if(0) {  
    cout << "Hello" ;  
}  
else  
{  
    cout << "Good Bye" ;  
}  
return 0;  
}
```

- a) Hello **b) Good Bye** c) HelloGood bye d) Compilation Error

4. if you have to make decision based on multiple choices, which of the following is best suited?

- a) if b) if-else **c) if-else-if** d) All of the above

5. Can we use string inside switch statement?

- a) Yes **b) No**

6. In situations where we need to execute body of the loop before testing the condition, we should use _____. a) For loop b) while loop **c) do-while loop** d) nested for loop

7. Loops in C++ Language are implemented using?

- a) While loop b) For loop c) Do while loop **d) All of the above**

8. While loop is faster in C++ Language, for, while or do-while?

- a) For b) while c) do-while **d) All work ate same speed**

9. What is the way to suddenly come out of or quit any loop in C++?

- a) continue; statement **b) break; statement**
c) leave; statement d) quit; statement

10. Which of the following can replace a simple if -wlsw construct?

- a) Ternary operator** b) while loop c) do-while loop d) fop loop

11. The if...else statement can be replaced by which operator?

- a) Bitwise operator **b) Conditional operator**
c) Multiplicative operator d) Addition operator

12. The switch statement is also called as?

- a) choosing structure **b) selective structure**
c) certain structure d) bitwise structure

13. Choose a correct C++ for loop syntax.

- a) for(initialization; condition; inc/dec){ // statements };**
b) for(declaration; condition; incrementoperation){ // statements };
c) for(declaration; incrementoperation; condition){ // statements };
d) for(initialization; condition; incrementoperation){ // statements };

14. Choose a correct C++ do while syntax.

- a) dowhile(condition){ // statements };
b) do while(condition){ // statements };
c) do{ // statements }while(condition)
d) do{ // statements }while(condition);

15. Choose a correct C++ Statement.

- a) a++ is (a= a+1) POST INCREMENT operator
b) a-- is (a = a-1) POST DECREMENT operator. --a is (a = a-1) PRE DECREMENT operator
c) ++a is (a = a+1) PRE INCREMENT operator
d) All of the above.

16. Which of the following is an entry-controlled loop?

- a) For loop b) while loop c) do-while loop **d) both b & c**

17. Which of the following is an exit-controlled loop?

- a) While loop b) For loop **c) Do while loop** d) both b & c

18. Which of the following is most suitable for a menu-driven program?

- a) For** b) while c) do-while d) All of the above

19. Consider the following loop: for(int i = 0; i<5; i++);

- a) It will give compilation error. **b) 5** c) 6 d) Some Garbage value

20. A switch construct can be used with which of the following types of variable?

- a) int **b) int, char** c) int, float, char d) Any basic datatype

21. The destination statement for the goto label is identified by what label?

- a) \$ b) @ c) * **d) :**

22. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n ;
    for (n = 5; n < 0; n--)
    {
        cout << n;
        if (n == 3)
            break;
    }
    return 0;
}
```

- a) 543** b) 54 c) 5432 d) 53

23. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int a = 10;
    if (a = 15)
    {
        time;
```



```
    cout << a;
    if (n == 3)
        goto time;
}
break;
return 0;
}
```

- a) 1010 b) 10 c) infinitely print 10 **d) compile time error**

24. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n = 15;
    for (; )
        cout << n;
    return 0;
}
```

- a) error b) 15 **c) infinite times of printing n** d) none of the mentioned

25. Which looping process is best used when the number of iterations is known?

- a) While loop **b) For loop** c) Do while loop
d) all looping processes require that the iterations be known

26. How many types of loops are there in C++?

- a) 4** b) 2 c) 3 d) 1

27. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int i;
    for (i = 0; i < 10; i++);
    {
        cout << i;
    }
    return 0;
}
```

- a) 0123456789 **b) 10** c) 012345678910 d) compile time error

28. Which of the following must be present in switch construct?

- a) Expression in () after switch** b) default
c) case followed by value d) All of the above

29. What is the effect of writing a break statement inside a loop?

- a) It cancels remaining iterations.** b) It skips a particular iteration.
c) The program terminates immediately. d) Loop counter is reset.

30. The break statement causes an exit

a) from the innermost loop only.

b) only from the innermost switch.

c) from all loops & switches.

d) from the innermost loop or switch.

Operators

1. Which operator is having the right to left associativity in the following?

a) Array subscripting

b) Function call

c) Addition and subtraction

d) Type cast

2. What are the essential operators in c++?

a) +

b) |

c) <=

d) All of the mentioned

3. In which direction does the assignment operation will take place?

a) left to right

b) right to left

c) top to bottom

d) bottom to top

4. Pick out the compound assignment statement.

a) a = a – 5

b) a = a / b

c) a -= 5

d) a = a + 5

5. What is the associativity of add(+); ?

a) right to left

b) left to right

c) right to left & left to right

d) top to bottom

6. What is the name of | operator?

a) sizeof

b) or

c) and

d) modulus

7. Which operator is having the highest precedence in c++?

a) array subscript

b) Scope resolution operator

c) static_cast

d) dynamic_cast

8. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    a = 10;
    b = 4;
    a = b;
    b = 7;
    cout << "a: " << a;
    cout << "\nb: " << b;
    return 0;
}
```

a) a:4 b:7

b) a:10 b:4

c) a:4 b:10

d) a:4 b:6

9. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int a, b, c;
    a = 2;
    b = 7;
    c = (a > b) ? a : b;
}
```



```
cout << "c: " << c;  
return 0;  
}
```

- a) 2 b) 7 c) 9 d) 14

10. What will be the output of the following C++ code?

```
#include <iostream>  
using namespace std;  
int main() {  
  
    int a = 0;  
    int b = 10;  
    a = 2;  
    b = 7;  
    if (a && b) {  
        cout << "true: " << endl;  
    }  
    else  
    {  
        cout << "false: " << endl;  
    }  
    return 0;  
}
```

- a) true b) false c) error d) 10

11. Which operator is having the highest precedence?

- a) postfix b) unary c) shift d) equality

12. What will be the output of the following C++ code?

```
#include <iostream>  
using namespace std;  
int main() {  
  
    int a;  
    a = 5 + 3 * 5;  
    cout << "a: " << a;  
  
    return 0;  
}
```

- a) 35 b) 20 c) 25 d) 30

13. What will be the output of the following C++ code?

```
#include <iostream>  
using namespace std;  
int main() {  
    int a = 5, b = 6, c, d;  
    c = a, b;  
    d = (a ,b);  
    cout << c << " " << d;  
    return 0;  
}
```


}

6 a) 5 5 b) 6 7 c) 6 8 d) 6

14. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int i, j;
    j = 10;
    i = (j++, j + 100, 999 + j);
    cout << i;
    return 0;
}
```

a) 1000 b) 11 c) 1010 d) 1001

15. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int x, y;
    x = 5;
    y = ++x * ++x;
    cout << x << y;
    y = x++ * ++x;
    cout << x << y;
    return 0;
}
```

a) 749735 b) 736749 c) 367497 d) 367597

16. What is this operator called "?:" ?

a) conditional b) relational c) casting operator d) unrelational

17. What is the use of dynamic_cast operator?

- a) it converts virtual base class to derived class
- b) it converts the virtual base object to derived objects
- c) it will convert the operator based on precedence
- d) it converts the virtual base object to derived class

18. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {

    int a = 5, b = 6, c;
    c = (a > b) ? a : b;
    cout << c;

    return 0;
}
```

a) 6 b) 5 c) 4 d) 7

19. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    double a = 21.09399;
    float b = 10.20;
    int c, d;
    c = (int) a;
    d = (int) b;
    cout << c << " " << d;
    return 0;
}
```

10 a) 20 21 b) 10 10 c) 21 20 d) 10

20. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int a = 20, b = 10, c = 15, d = 5;
    int e;
    e = a + b * c / d;
    cout << e << endl;
    return 0;
}
```

a) 50 b) 60 c) 70 d) 90

C++ Functions

1. Where does the execution of the program starts?

- a) user-defined function **b) main function**
c) void function d) else function

2. What are mandatory parts in the function declaration?

- a) return type, function name** b) return type, function name, parameters
c) parameters, function name d) parameters, variables

3. which of the following is used to terminate the function declaration?

- a) : b)) **c) ;** d)]

4. How many can max number of arguments present in function in the c99 compiler?

- a) 99 b) 90 c) 102 **d) 127**

5. Which is more effective while calling the functions?

- a) call by value **b) call by reference** c) call by pointer d) call by object

6. What will be the output of the following C++ code?

```
#include <iostream>
```



```
using namespace std;
void car{
    cout << "Audi R8";
}
int main() {
    car();
    return 0;
}
```

- a) Audi R8 b) Audi R8Audi R8 c) **compile time error** d) runtime error

7. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
void fun(int x, int y){
    x = 20;
    y = 10;
}
int main() {
    int x = 10;
    fun (x, x);
    cout << x;
    return 0;
}
```

- a) **10** b) 20 c) compile time error d) 30

8. What happens to a function defined inside a class without any complex operations (like looping, a large number of lines, etc)?

- a) It becomes a virtual function of the class b) It becomes a default calling function of the class
c) **It becomes an inline function of the class** d) The program gives an error

9. What is the scope of the variable declared in the user defined function?

- a) whole program b) **only inside the {} block** c) the main function d) header section

10. How many minimum number of functions should be present in a C++ program for its execution?

- a) 0 b) **1** c) 2 d) 3

11. Which of the following is the default return value of functions in C++?

- a) **int** b) char c) float d) void

12. What is an inline function?

- a) **A function that is expanded at each call during execution**
b) A function that is called during compile time
c) A function that is not checked for syntax errors
d) A function that is not checked for semantic analysis

13. An inline function is expanded during _____

- a) **compile-time** b) run-time c) never expanded d) end of the program

14. In which of the following cases inline functions may not work?

- i) If the function has static variables. ii) If the function has global and register variables.
iii) If the function contains loops iv) If the function is recursive
a) i, iv b) iii, iv c) ii, iii, iv **d) i, iii, iv**

15. When we define the default values for a function?

- a) When a function is defined **b) When a function is declared**
c) When the scope of the function is over d) When a function is called

16. Where should default parameters appear in a function prototype?

- a) To the rightmost side of the parameter list** b) To the leftmost side of the parameter list
c) Anywhere inside the parameter list d) Middle of the parameter list

17. If an argument from the parameter list of a function is defined constant then _____

- a) It can be modified inside the function **b) It cannot be modified inside the function**
c) Error occurs d) Segmentation fault

18. Which of the following feature is used in function overloading and function with default argument? a) Encapsulation **b) Polymorphism** c) Abstraction d) Modularity

19. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int fun(int x = 0, int y = 0, int z){
    return (x + y + z);
}
int main() {
    cout << fun(10);
    return 0;
}
```

- a) 10 b) 0 **c) Error** d) Segmentation fault

20. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int fun(int = 0, int = 0);
int main() {
    cout << fun(5);
    return 0;
}
int fun(int x, int y){
    return (x + y);
}
```

- a) -5 b) 0 c) 10 **d) 5**

C++ Arrays & Strings

1. Which of the following correctly declares an array?

- a) int array[10];** b) int array; c) array{10}; d) array array[10];

2. What is the index number of the last element of an array with 9 elements?

- a) 9 b) 8 c) 0 d) Programmer-defined

3. What is the correct definition of an array?

- a) **An array is a series of elements of the same type in contiguous memory locations**
b) An array is a series of element
c) An array is a series of elements of the same type placed in non-contiguous memory locations
d) An array is an element of the different type

4. Which of the following gives the memory address of the first element in array?

- a) array[0]; b) array[1]; c) array(2); d) **array;**

5. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int array1[] = {1200, 200, 2300, 1230, 1543};
int array2[] = {12, 14, 16, 18, 20};
int temp, result = 0;
int main() {
    for (temp = 0; temp < 5; temp++){
        result += array1[temp];
    }
    for (temp = 0; temp < 4; temp++){
        result += array2[temp];
    }
    cout << result;
    return 0;
}
```

- a) 6553 b) **6533** c) 6522 d) 12200

6. Which of the following accesses the seventh element stored in array?

- a) **array[6];** b) array[7]; c) array(7); d) array;

7. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int array[] = {0, 2, 4, 6, 7, 5, 3};
    int n, result = 0;
    for (n = 0; n < 8; n++){
        result += array[n];
    }
    cout << result;
    return 0;
}
```

- a) 25 b) 26 c) **27** d) 21

8. What will be the output of the following C++ code?

```
#include <iostream>
```



```
using namespace std;
int main() {
    int a = 5, b = 10, c = 15;
    int arr[3] = {&a, &b, &c};
    cout << *arr[*arr[1] - 0];
    return 0;
}
```

- a) 15 b) 18 c) garbage value **d) compile time error**

9. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    char str[5] = "ABC";
    cout << str [3];
    cout << str;
    return 0;
}
```

- a) ABC** b) ABCD c) AB d) AC

10. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    char array[] = [10, 20, 30];
    cout << -2 [array];
    return 0;
}
```

- a) -15 **b) -30** c) compile time error d) garbage value

11. What is the difference between unsigned int length() and unsigned int size()?

- a) Returns a different value **b) They are same**
c) Returns a different value but they are same d) Returns a length

12. Which header file is used to manipulate the string?

- a) iostream b) iomanip **c) string** d) container

13. How many maximum number of parameters does a string constructor can take?

- a) 1 b) 2 **c) 3** d) 4

14. Which constant member functions does not modify the string?

- a) bool empty()** b) assign c) append d) delete

15. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    string str {"Steve jobs"};
```



```
cout << str.capacity() << "\n";
return 0;
}
```

- a) 9 b) 10 c) 11 d) Not Fix

16. Wht will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    string str {"Steve jobs founded the apple"};
    string str {"apple"};
    unsigned found = str.find(str2);
    if (found != string :: npos)
        cout << found << "\n";
    return 0;
}
```

- a) apple b) 12 c) 23 d) Steve jobs founded the

17. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    string str {"Steve jobs"};
    unsigned long int found = str.find_first_of("aeiou");
    while (found != string :: npos)
    {
        str[found] = '*';
        found = str.find_first_of("aeiou", found + 1);
    }
    cout << str << "\n";
    return 0;
}
```

- a) Steve b) jobs c) St*v* j*bs d) St*v*

18. What will be the output of the following C++ code?

```
#include <iostream>
#include <cstring>
#include <string>
using namespace std;
int main() {
    string str {"Steve jobs"};
    char * cstr = new char [str.length() + 1];
    strcpy (cstr, str.c_str());
    char * p = strtok (cstr, " ");
    while (p != 0)
    {
        cout << p << "\n";
        p = strtok(NULL, " ");
    }
}
```



```
}  
delete[] cstr;  
  
return 0;  
}
```

- a) Steve jo b) Steve jobs c) Steve jobs d) Steve jo

19. How many parameters can a resize method take?

- a) 1 b) 2 c) 1 or 2 d) 2

20. Which of the following is correct about remove_extent() function?

- a) Removes the given dimension from an array
b) Removes the first dimension from the right of the array
c) Removes the first dimension from the left of the array
d) Removes the last dimension from the left of the array

21. Which of the header file is used for array type manipulation?

- a) <array> b) <type_traits> c) <iostream> d) std namespace

22. What is the use of is_array() function in C++?

- a) To check if a variable is array type or not** b) To check if a variable is 1-D array type or not
c) To check if a variable is 2-D array type or not
d) To check if a variable is 1-D or 2-D array type or not

23. What is the use of is_same() function in C++?

- a) To check if a variable is array type or not
b) To check whether two variables have the same characteristics
c) To check if two variable is of array type or not
d) To check whether two variables are different or not

24. Which of the following is correct about extent() function?

- a) Returns how many elements are in array currently
b) Returns the size of the 1st dimension
c) Returns how many total elements can be stored in an array
d) Returns the size of a given dimension

25. What will be the output of the following C++ code?

```
#include <iostream>  
#include <string>  
using namespace std;  
int main() {  
    cout << is_same <int, char> :: value;  
    cout << is_same <char [10], char [10]> :: value;  
    cout << is_same <char * [10], string> :: value;  
    return 0;  
}
```

- a) 011 b) 101 c) 010 d) 110

26. What will be the output of the following C++ code?


```
#include <iostream>
#include <string>
using namespace std;
int main() {
    cout << rank <int [10]> :: value;
    cout << rank <char [10] [10]> :: value;
    cout << rank < string [10] [10] [10]> :: value;
    return 0;
}
```

- a) 111 b) 123 c) 321 d) 121

27. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    cout << is_array <int> :: value;
    cout << is_array <char [10]> :: value;
    cout << is_array <string> :: value;
    return 0;
}
```

- a) 010 b) 101 c) 001 d) 110

28. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    cout << extent < string [10][20][30], 0> :: value;
    cout << extent < string [10][20][30], 1> :: value;
    cout << extent < string [10][20][30], 2> :: value;
    return 0;
}
```

- a) 101010 b) 102030 c) 302010 d) 102010

29. What is the use of rank() function in C++?

- a) Returns size of each dimension b) Returns how many total elements can be stored in an array
c) Returns how many elements are in array currently d) **Returns the dimension of an array**

30. Which of the following is correct about remove_all_extents() function?

- a) **Removes the all dimension from an array**
b) Removes the first dimension from the left of the array
c) Removes the first dimension from the right of the array
d) Removes the last dimension from the left of the array

Structures

1. What will be the output of the following C++ code?

```
#include <iostream>
```



```
#include <string.h>
using namespace std;
int main() {
    int student{
        int num;
        char name[25];
    }
    student stu;
    stu.num = 123;
    strcpy(stu.name, "john");
    cout << stu.num << endl;
    cout << stu.name << endl;
    return 0;
}
```

- a) 123 john b) john john c) compile time error d) runtime error

2. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
struct Time{
    int hours, minutes, seconds;
}
int toSeconds(Time now);
int main() {
    Time t;
    t.hours = 5;
    t.minutes = 30;
    t.seconds = 45;
    cout << "Total seconds: " << toSeconds(t) << endl;
    return 0;
}
int toSeconds (Time now){
    return 3600 * now.hours + 60 * now.minutes + now.seconds;
}
```

- a) 19845 b) 20000 c) 15000 d) 19844

3. What will happen when the structure is declared?

- a) it will not allocate any memory b) it will allocate the memory
c) it will be declared and initialized d) it will be declared

4. The declaration of the structure is also called as?

- a) structure creator b) structure signifier
c) **structure specifier** d) structure creator & signifier

5. The data elements in the structure are also known as what?

- a) objects b) **members** c) data d) objects & data

6. What will be used when terminating a structure?

- a) : b) } c) ; d) ;;

7. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    struct ShoeType{
        string style;
        double price;
    };
    ShoeType shoe1, shoe2;
    shoe1.style = "Adidas";
    shoe1.price = 9.99;
    cout << shoe1.style << "$" << shoe1.price;
    shoe2 = shoe1;
    shoe2.price = shoe2.price / 9;
    cout << shoe2.style << "$" << shoe2.price;
    return 0;
}
```

- a) Adidas \$ 9.99Adidas \$ 1.11 b) Adidas \$ 9.99Adidas \$ 9.11
c) Adidas \$ 9.99Adidas \$ 11.11 d) Adidas \$ 11.11Adidas \$ 11.11

8. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
struct sec{
    int a;
    char b;
};
int toSeconds(Time now);
int main() {
    struct sec s = {25, 50};
    struct sec *ps = (struct sec *) &s;
    cout << ps->a << ps->b ;
    return 0;
}
```

- a) 252 b) 253 c) 254 d) 262

9. Which of the following is a properly defined structure?

- a) struct {int a;} b) struct a_struct {int a;}
c) struct a_struct int a; d) **struct a_struct {int a;};**

10. Which of the following accesses a variable in structure *b?

- a) **b->var;** b) b.var; c) b-var; d) b>var;

C++ Pointers

1. What does the following statement mean?

```
int (*fp) (char*);
```



```
cout << arr;
return 0;
}
```

- a) ABCDEFGHIJ b) AAAAAAAAAA c) JJJJJJJ d) AAAAAAJJJ

10. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    char *ptr;
    char str[] = 'abcdefg';
    ptr = str;
    ptr += 5;
    cout << ptr;
    return 0;
}
```

- a) fg b) cdef c) defg d) abcd

11. What is the meaning of the following declaration?

```
int (*p [5]) ();
```

- a) p is pointer to function **b) p is array of pointer to function**
c) p is pointer to such function which return type is the array d) p is pointer to array of function

12. What is size of generic pointer in C++ (in 32-bit platform)?

- a) 2 **b) 4** c) 8 d) 0

13. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int a[2][4] = {3, 6, 9, 12, 15, 18, 21, 24};
    cout << *(a[1] + 2) << * (*(a + 1) + 2) << [1[a]];
    return 0;
}
```

- a) 15 18 21 **b) 21 21 21** c) 24 24 24 d) Compile time error

14. When does the void pointer can be dereferenced?

- a) when it doesn't point to any value **b) when it cast to another type of object**
c) using delete keyword d) using shift keyword

15. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int arr[] = {4, 5, 6, 7};
    int *p = (arr + 1);
    cout << *p;
    return 0;
}
```


}

- a) 4 b) 5 c) 6 d) 7

16. What will happen in the following C++ code snippet?

```
#include <iostream>
using namespace std;
int main() {
    int arr[] = {4, 5, 6, 7};
    int *p = (arr + 1);
    cout << arr;
    return 0;
}
```

- a) 4 b) 5 c) address of arr d) 7

17. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int numbers[5];
    int *p;
    p = numbers; *p = 10;
    p++; *p = 20;
    p = numbers[2]; *p = 30;
    p = numbers + 3; *p = 40;
    p = numbers; *(p + 4) = 50;
    for (int n = 0; n < 5; n++){
        cout << numbers[n] << ",";
    }
    return 0;
}
```

- a) 10,20,30,40,50, b) 1020304050 c) compile error d) runtime error

18. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int arr[] = {4, 5, 6, 7};
    int *p = (arr + 1);
    cout << *arr + 9;
    return 0;
}
```

- a) 12 b) 5 c) 13 d) error

19. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int i;
    const char *arr[] = {"c", "c++", "java", "VBA"};
```



```
const char *(*ptr)[4] = &arr;  
cout << ++(*ptr)[2];  
return 0;  
}
```

- a) **ava** b) java c) c++ d) compile time error

20. The void pointer can point to which type of objects?

- a) int b) float c) double d) **all of the mentioned**

C++ File Management

1. Which header file is required to use file I/O operations?

- a) <ifstream> b) <ostream> c) **<fstream>** d) <iostream>

2. Which stream class is to only write on files?

- a) **ofstream** b) ifstream c) istream d) fstream

3. Which of the following is used to create an output stream?

- a) **ofstream** b) ifstream c) istream d) fstream

4. Which of the following is used to create a stream that performs both input and output operations?

- a) ofstream b) ifstream c) istream d) **fstream**

5. Which of the following is not used as a file opening mode?

- a) **ios::trunc** b) ios::binary c) ios::in d) ios::ate

6. Which of the following statements are correct?

- 1) It is not possible to combine two or more file opening mode in open() method.
- 2) It is possible to combine two or more file opening mode in open() method.
- 3) ios::in and ios::out are input and output file opening mode respectively.

- a) **1, 3** b) 2, 3 c) 3 only d) 1, 2

7. By default, all the files in C++ are opened in _____ mode.

- a) **Text** b) Binary c) ISCHII d) VTC

8. What is the use of ios::trunc mode?

- a) To open a file in input mode b) To open a file in output mode
c) To truncate an existing file to half d) **To truncate an existing file to zero**

9. Which of the following is the default mode of the opening using the ofstream class?

- a) ios::in b) **ios::out** c) ios::app d) ios::trunc

10. What is the return type open() method?

- a) int b) char c) **bool** d) float

11. Which of the following is not used to seek file pointer?

- a) **ios::set** b) ios::end c) ios::cur d) ios::beg

12. Which of the following is the default mode of the opening using the ifstream class?

- a) **ios::in** b) ios::out c) ios::app d) ios::trunc

13. Which of the following is the default mode of the opening using the fstream class?

- a) ios::in b) ios::out **c) ios::in|ios::out** d) ios::trunc

14. Which function is used in C++ to get the current position of file pointer in a file?

- a) **tell_p()** b) get_pos() c) get_p() d) tell_pos()

15. Which function is used to reposition the file pointer?

- a) moveg() **b) seekg()** c) changep() d) go_p()

16. Which of the following is used to move the file pointer to start of a file?

- a) **ios::beg** b) ios::start c) ios::cur d) ios::first

17. It is not possible to combine two or more file opening mode in open () method.

- a) True b) False c) May be **d) None of these**

18. Which of these is the correct statement about eof() ?

- a) Returns true if a file open for reading has reached the next character.
b) Returns true if a file open for reading has reached the next word.
c) Returns true if a file open for reading has reached the end.
d) Returns true if a file open for reading has reached the middle.

19. Which of the following true about FILE *fp

- a) FILE is a structure and fp is a pointer to the structure of FILE type**
b) FILE is a buffered stream c) FILE is a stream
d) FILE is a keyword in C for representing files and fp is a variable of FILE type

20. Which of the following methods can be used to open a file in file handling?

- a) Using Open () b) Constructor method c) Destructor method **d) Both A and B**

21. Which operator is used to insert the data into file?

- a) >> **b) <<** c) < d) None of the above

22. Which is correct syntax?

- a) myfile:open ("example.bin", ios::out); **b) myfile.open ("example.bin", ios::out);**
c) myfile::open ("example.bin", ios::out); d) myfile.open ("example.bin", ios:out);

23. In fopen(), the open mode "wx" is sometimes preferred "w" because. i) Use of wx is more efficient. ii) If w is used, old contents of file are erased and a new empty file is created. When wx is used, fopen() returns NULL if file already exists.

- a) Only I **b) Only ii** c) Both i & ii d) None of the above

24. ios::trunc is used for ?

- a) If the file is opened for output operations and it already existed, no action is taken.
b) If the file is opened for output operations and it already existed, then a new copy is created.
c) None of the above

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d) If the file is opened for output operations and it already existed, its previous content is deleted and replaced by the new one.

25. Which member function is used to determine whether the stream object is currently associated with a file? **a) is_open** b) Buf c) String d) None of the above

26. getc() returns EOF when
a) End of files is reached b) When getc() fails to read a character
c) Both A & B d) None of the above

27. Which of the following is not a file opening mode _____.
a) ios::ate b) ios::nocreate c) ios::noreplace **d) ios::truncate**

28. Due to ios::trunc mode, the file is truncated to zero length.
a) True b) False

29. If we have object from fstream class, then what is the default mode of opening the file?
a) ios::in|ios::out b) ios::in|ios::out|ios::trunk
c) ios::in|ios::trunk **d) Default mode depends on compiler**

30. What is the output of this program?

```
#include <iostream>
#include <string>
using namespace std;
int main() {
    char fine, course;
    cout << "Enter a word: ";
    fine = cin.get();
    cin.sync();
    course = cin.get();
    cout << fine << endl;
    cout << course << endl;
    return 0;
}
```

a) course b) fine c) None of the mentioned
d) Returns fine 2 letter or number from the entered word

